MECHANICS OF MATERIALS

Lecture 23: Plane Problems in Linear Elasticity I

Ravi Sastri Ayyagari

Mechanical Engineering
Indian Institute of Technology Gandhinagar

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Plane Problems

1. **Plane Stress** : If a thin plate is loaded by forces that are applied at the boundary, parallel to the plane of the plate and distributed uniformly over the thickness the out of plane components are zero.
   Typically, $\sigma_{xz} = \sigma_{yz} = \sigma_{zz} = 0$, in this case .
2. **Plane Strain:** When dimensions are very large along one direction (say, z) in comparison to the other two orthogonal directions (x and y).
   Typically, $\epsilon_{xz} = \epsilon_{yz} = \epsilon_{zz} = 0$, in this case.

Plane Stress - Equations

1. Under plane stress conditions : $\sigma_{xz} = \sigma_{yz} = \sigma_{zz} = 0$
2. Strain-displacement relations :

$$\epsilon_{xx} = \frac{\partial u}{\partial x}; \quad \epsilon_{yy} = \frac{\partial v}{\partial y}; \quad \epsilon_{zz} = \frac{\partial w}{\partial z}; \quad \epsilon_{xy} = \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)$$

3. Equilibrium equations :

$$\sigma_{xx,x} + \sigma_{xy,x} + f_x = \rho\ddot{u}$$
$$\sigma_{xy,y} + \sigma_{yy,y} + f_y = \rho\ddot{v}$$

Note : f_x and f_y are body forces - that are typically considered as zero. $\ddot{u}$, $\ddot{v}$ are the accelerations which in case of quasi-static conditions can be ignored.

Plane Stress - Equations

1. Generalized Hooke's Law (stresses in terms of strains):

$$\sigma_{xx} = \frac{E}{(1-\nu^2)}(\epsilon_{xx} + \nu\epsilon_{yy}); \quad \sigma_{yy} = \frac{E}{(1-\nu^2)}(\epsilon_{yy} + \nu\epsilon_{xx}); \quad \sigma_{xy} = \frac{E}{(1+\nu)}\epsilon_{xy}$$

2. Generalized Hooke's Law (strains in terms of stresses):

$$\epsilon_{xx} = \frac{1}{E}[\sigma_{xx} - \nu\sigma_{yy}]; \quad \epsilon_{xy} = \frac{1}{2G}\sigma_{xy};$$
$$\epsilon_{yy} = \frac{1}{E}[\sigma_{yy} - \nu\sigma_{xx}]; \quad \epsilon_{xz} = 0$$
$$\epsilon_{zz} = -\frac{\nu}{E}[\sigma_{xx} + \sigma_{yy}]; \quad \epsilon_{yz} = 0$$

Plane Stress - Equations

1. Compatibility equations:

$$\epsilon_{xx,yy} + \epsilon_{yy,xx} = 2\epsilon_{xy,xy}$$
$$\epsilon_{yy,zz} + \epsilon_{zz,yy} = 0$$
$$\epsilon_{zz,xx} + \epsilon_{xx,zz} = 0$$
$$\epsilon_{zz,xy} + \epsilon_{xy,zz} = 0$$
$$\epsilon_{yy,xz} = \epsilon_{xy,yz}$$
$$\epsilon_{xx,yz} = \epsilon_{xy,xz}$$

2. Invariant relations :

$$\sigma_{xx} + \sigma_{yy} = \frac{E}{1-\nu}(\epsilon_{xx} + \epsilon_{yy}); \quad \epsilon_{xx} + \epsilon_{yy} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$$

Plane Strain - Equations

1. Under plane strain conditions : $\epsilon_{xz} = \epsilon_{yz} = \epsilon_{zz} = 0$
2. Strain-displacement relations :

$$\epsilon_{xx} = \frac{\partial u}{\partial x}; \quad \epsilon_{yy} = \frac{\partial v}{\partial y}; \quad \epsilon_{xy} = \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)$$

3. Equilibrium equations :

$$\sigma_{xx,x} + \sigma_{xy,y} = 0; \quad \sigma_{xy,x} + \sigma_{yy,y} = 0; \quad \sigma_{zz,z} = 0$$

4. Generalized Hooke's Law :

$$\epsilon_{xx} = \frac{1}{E^\star}\left(\sigma_{xx} - \nu^\star\sigma_{yy}\right); \quad \epsilon_{yy} = \frac{1}{E^\star}\left(\sigma_{yy} - \nu^\star\sigma_{xx}\right); \quad \epsilon_{xy} = \frac{\sigma_{xy}}{2G}$$

where:

$$E^\star = \frac{E}{1-\nu^2}; \quad \nu^\star = \frac{\nu}{1-\nu}; \quad G = \frac{E}{2(1+\nu)}$$

Plane Strain - Equations

1. Stresses :

$$\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy}); \quad \sigma_{zx} = \sigma_{zy} = 0$$

2. Generalized Hooke's Law :

$$\epsilon_{xx} = \frac{1}{E}[(1-\nu^2)\sigma_x - \nu(1+\nu)\sigma_y]$$
$$\epsilon_{yy} = \frac{1}{E}[(1-\nu^2)\sigma_y - \nu(1+\nu)\sigma_x]$$
$$\gamma_{xy} = \frac{2(1+\nu)}{E}\tau_{xy}$$

3. Compatibility:

$$\epsilon_{xx,yy} + \epsilon_{yy,xx} = 2\epsilon_{xy,xy}$$

Boundary conditions

1. Displacement boundary conditions : displacement specified on entire boundary.
2. Traction boundary conditions : traction specified on entire boundary.
3. Mixed boundary conditions: displacement specified on part of the boundary while tractions are specified on part boundary.
4. Robin boundary conditions : both tractions and displacements are specified on the same part of boundary

Types of Boundary value problems

1. Displacement Type BVP
   Here displacements are prescribed.
   - Step 1: Write the equilibrium equations.
   - Step 2: Recast equilibrium equations in terms of strains using stress-strain relations
   - Step 3: Recast the resulting equation in Step 2 in terms of displacements using strain-displacement relations. This results in Navier's Equation.
2. Traction Type BVP
   Here tractions are prescribed.
   - Step 1: Write the compatibility equations.
   - Step 2: Recast compatibility equations in terms of stresses using stress-strain relations
   - Step 3: Simplify the resulting equation in Step 2 using equilibrium equations to reduce the equation to an equation that is in terms of stresses. This results in Beltrami Mitchell Equation.

Stress Method

Plane stress case :

- Step 1: Use compatibility equation and recast it using Hooke's Law in terms of stresses:

$$\frac{\partial^2 \epsilon_{xx}}{\partial y^2} + \frac{\partial^2 \epsilon_{yy}}{\partial x^2} = 2\frac{\partial^2 \epsilon_{xy}}{\partial x \partial y}$$
$$\frac{\partial^2}{\partial y^2}\left(\sigma_{xx} - \nu\sigma_{yy}\right) + \frac{\partial^2}{\partial x^2}\left(\sigma_{yy} - \nu\sigma_{xx}\right) = 2(1+\nu)\frac{\partial^2 \tau_{xy}}{\partial x \partial y}$$

- Step 2: From equilibrium (in absence of body forces and under quasi-static conditions), we have :

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0; \quad \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = 0$$

Stress Method

Plane stress case :

- Step 3: Taking partial differential of the first equation with respect to x and second with respect to y and adding we obtain:

$$2\frac{\partial^2 \tau_{xy}}{\partial x \partial y} = -\frac{\partial^2 \sigma_{xx}}{\partial x^2} - \frac{\partial^2 \sigma_{yy}}{\partial y^2}$$

- Step 4: Substituting above equation in the compatibility relation in Step 1 we get :

$$\frac{\partial^2}{\partial y^2}\left(\sigma_{xx} - \nu\sigma_{yy}\right) + \frac{\partial^2}{\partial x^2}\left(\sigma_{yy} - \nu\sigma_{xx}\right) = (1+\nu)\left(-\frac{\partial^2 \sigma_{xx}}{\partial x^2} - \frac{\partial^2 \sigma_{yy}}{\partial y^2}\right)$$
$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\left(\sigma_{xx} + \sigma_{yy}\right) = 0$$
$$\nabla^2(\sigma_{xx} + \sigma_{yy}) = 0$$

Stress Method

Plane stress case :

- Step 5 : Any permissible stress field has to satisfy the Laplace equation (in Step 4) in addition to the equilibrium equations in Step 2. Let us therefore consider:

$$\sigma_{xx} = \frac{\partial^2 \phi}{\partial y^2} = \phi_{,yy}; \quad \sigma_{yy} = \frac{\partial^2 \phi}{\partial x^2} = \phi_{,xx}; \quad \tau_{xy} = -\frac{\partial^2 \phi}{\partial x \partial y} = -\phi_{,xy}$$

Then:

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0; \quad \Longrightarrow \quad \frac{\partial^3 \phi}{\partial x \partial y^2} - \frac{\partial^3 \phi}{\partial x \partial y^2} = 0 \quad \text{(satisfied)}$$
$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = 0; \quad \Longrightarrow \quad -\frac{\partial^3 \phi}{\partial x^2 \partial y} + \frac{\partial^3 \phi}{\partial x^2 \partial y} = 0 \quad \text{(satisfied)}$$
$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\left(\sigma_{xx} + \sigma_{yy}\right) = 0; \quad \Longrightarrow \quad \frac{\partial^4 \phi}{\partial x^4} + 2\frac{\partial^4 \phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \phi}{\partial y^4} = 0$$

Stress Method

Plane stress case :

- Step 6 : The governing equation in ϕ can be written as:

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)(\sigma_{xx}+\sigma_{yy})=0; \quad \Longrightarrow \quad \frac{\partial^4\phi}{\partial x^4}+2\frac{\partial^4\phi}{\partial x^2\partial y^2}+\frac{\partial^4\phi}{\partial y^4}=0$$
$$\Longrightarrow \quad \nabla^2(\nabla^2\phi)=0$$
$$\Longrightarrow \quad \nabla^4\phi=0$$

where: ϕ is the *Airy stress function* and the governing equation in case of plane stress is a *biharmonic equation*.

Objective : Determine ϕ.

Stress Method

Plane strain case :

- Step 1: Use compatibility equation and recast it using Hooke's Law in terms of stresses:

$$\frac{\partial^2\epsilon_{xx}}{\partial y^2}+\frac{\partial^2\epsilon_{yy}}{\partial x^2}=2\frac{\partial^2\epsilon_{xy}}{\partial x\partial y}$$
$$\frac{\partial^2}{\partial y^2}\left((1-\nu^2)\sigma_{xx}-\nu(1+\nu)\sigma_{yy}\right)+\frac{\partial^2}{\partial x^2}\left((1-\nu^2)\sigma_{yy}-\nu(1+\nu)\sigma_{xx}\right)=2(1+\nu)\frac{\partial^2\tau_{xy}}{\partial x\partial y}$$
$$\frac{\partial^2}{\partial y^2}\left((1-\nu)\sigma_{xx}-\nu\sigma_{yy}\right)+\frac{\partial^2}{\partial x^2}\left((1-\nu)\sigma_{yy}-\nu\sigma_{xx}\right)=2\frac{\partial^2\tau_{xy}}{\partial x\partial y}$$

- Step 2: From equilibrium (in absence of body forces and under quasi-static conditions), we have :

$$\frac{\partial\sigma_{xx}}{\partial x}+\frac{\partial\tau_{xy}}{\partial y}=0; \quad \frac{\partial\tau_{xy}}{\partial x}+\frac{\partial\sigma_{yy}}{\partial y}=0; \quad \frac{\partial\sigma_{zz}}{\partial z}=0$$

Stress Method

Plane strain case :

- Step 3: Taking partial differential of the first equation with respect to x and second with respect to y and sdding we obtain:

$$2\frac{\partial^2\tau_{xy}}{\partial x\partial y}=-\frac{\partial^2\sigma_{xx}}{\partial x^2}-\frac{\partial^2\sigma_{yy}}{\partial y^2}$$

- Step 4: Substituting above equation in the compatibility relation in Step 1 we get :

$$\frac{\partial^2}{\partial y^2}\left((1-\nu)\sigma_{xx}-\nu\sigma_{yy}\right)+\frac{\partial^2}{\partial x^2}\left((1-\nu)\sigma_{yy}-\nu\sigma_{xx}\right)=\left(-\frac{\partial^2\sigma_{xx}}{\partial x^2}-\frac{\partial^2\sigma_{yy}}{\partial y^2}\right)$$
$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)(\sigma_{xx}+\sigma_{yy})=0$$
$$\nabla^2(\sigma_{xx}+\sigma_{yy})=0$$

Stress Method

Plane strain case :

- Step 5 : Any permissible stress field has to satisfy the Laplace equation (in Step 4) in addition to the equilibrium equations in Step 2. Let us therefore consider:

$$\sigma_{xx}=\frac{\partial^2\phi}{\partial y^2}=\phi_{,yy}; \quad \sigma_{yy}=\frac{\partial^2\phi}{\partial x^2}=\phi_{,xx}; \quad \tau_{xy}=-\frac{\partial^2\phi}{\partial x\partial y}=-\phi_{,xy}$$

Then:

$$\frac{\partial\sigma_{xx}}{\partial x}+\frac{\partial\tau_{xy}}{\partial y}=0; \quad \Longrightarrow \quad \frac{\partial^3\phi}{\partial x\partial y^2}-\frac{\partial^3\phi}{\partial x\partial y^2}=0 \quad \text{(satisfied)}$$
$$\frac{\partial\tau_{xy}}{\partial x}+\frac{\partial\sigma_{yy}}{\partial y}=0; \quad \Longrightarrow \quad -\frac{\partial^3\phi}{\partial x^2\partial y}+\frac{\partial^3\phi}{\partial x^2\partial y}=0 \quad \text{(satisfied)}$$
$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)(\sigma_{xx}+\sigma_{yy})=0; \quad \Longrightarrow \quad \frac{\partial^4\phi}{\partial x^4}+2\frac{\partial^4\phi}{\partial x^2\partial y^2}+\frac{\partial^4\phi}{\partial y^4}=0$$

Stress Method

Plane strain case :

- Step 6 : The governing equation in ϕ can be written as:

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)(\sigma_{xx}+\sigma_{yy})=0; \quad \Longrightarrow \quad \frac{\partial^4\phi}{\partial x^4}+2\frac{\partial^4\phi}{\partial x^2\partial y^2}+\frac{\partial^4\phi}{\partial y^4}=0$$
$$\Longrightarrow \quad \nabla^2(\nabla^2\phi)=0$$
$$\Longrightarrow \quad \nabla^4\phi=0$$

where: ϕ is the *Airy stress function* and the governing equation in case of plane strain is also the same *biharmonic equation* as in case of plane stress.

Objective : Determine ϕ.

Solutions to Biharmonic Equation

1. In case of plane static problems when body forces are neglected the governing equation is:

$$\nabla^4\phi=0; \quad \sigma_{xx}=\frac{\partial^2\phi}{\partial y^2}; \quad \sigma_{yy}=\frac{\partial^2\phi}{\partial x^2}; \quad \sigma_{xy}=\frac{\partial^2\phi}{\partial x\partial y}$$

2. Broadly there are two methods of solving the *biharmonic equation* :
   - Inverse Methods: We examine solution and identify BVP corresponding to it.
   - Semi-inverse method: Stress or displacement distribution is assumed.
     - Polynomial method
     - Fourier series method
     - Complex variable method

We will limit the plane problem analysis to *Polynomial Method* only.

Working Assignment

Work out the plane stress and plane strain equations